Autonomous Compatibility Standard (ACPS)

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Governed Space

Autonomous Operational Compatibility

Core Question

How do autonomous systems remain operationally compatible?

Canonical Definition

Autonomous Compatibility Standard (ACPS) defines the structural conditions under which autonomous operational systems remain compatible, interoperable, governable, and operationally aligned while interacting across heterogeneous autonomous operational environments throughout the complete autonomous operational lifecycle.

ACPS governs autonomous operational compatibility.

The standard establishes the governance conditions required to ensure that autonomous operational actors remain capable of operating together despite differences in technologies, platforms, vendors, operational environments, and governance implementations.

Autonomous systems rarely operate alone.

Compatibility enables autonomous ecosystems.

ACPS governs that compatibility.

Governance Flow Position

Autonomous Identity → Autonomous Authority → Autonomous Execution → Autonomous Constraints → Autonomous Risk & Trust → Autonomous Coordination → Autonomous Escalation → Autonomous Operational Accountability → **Autonomous Compatibility** → Autonomous Evolution

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Standard Abstract

Future autonomous environments will consist of diverse autonomous systems developed by different organizations using different technologies.

Without governed compatibility:

- operational integration becomes unreliable,
- coordination weakens,
- interoperability decreases,
- governance consistency is lost,
- autonomous ecosystems become fragmented.

ACPS establishes the governance conditions required for trustworthy autonomous compatibility across complex operational ecosystems.

Operational Role

Autonomous Compatibility Standard provides the governance layer responsible for preserving operational compatibility across autonomous operational actors, platforms, and ecosystems.

Its role is to ensure that autonomous systems remain capable of cooperating safely, consistently, and under shared governance conditions.

Operational Space

ACPS governs:

- operational interoperability,
- protocol compatibility,
- cross-system trust,
- governance compatibility,
- compatibility validation,
- operational integration,

- ecosystem compatibility,
- governed interoperability.

This governed space ensures that autonomous operational ecosystems remain compatible despite technological diversity.

Why This Standard Exists

Autonomous ecosystems depend on collaboration between independently developed operational actors.

Governance must ensure that compatibility is preserved throughout autonomous interaction.

ACPS exists because autonomous operational compatibility represents the **ninth governable space** of **Autonomous Systems Governance Architecture (ASGA™).**

Scope

This standard may be applied across:

- multi-agent AI systems,
- robotics,
- autonomous vehicles,
- industrial automation,
- healthcare robotics,
- enterprise AI platforms,
- critical infrastructure,
- defense autonomy,
- Human-AI operational environments,
- distributed autonomous ecosystems.

Relationship to Other OOF® Architectures

ACPS governs autonomous operational compatibility.

It interoperates with:

- **GOA™** for governance principles,
- **ORA™** for operational reality,
- **AGA™** for accountability,
- **AIG®** for AI behavioral governance,
- **CLIA®** for cognition,
- **MGIA™** for memory governance.

Together these architectures establish the governance foundation required for compatible, interoperable, and trustworthy autonomous operational ecosystems.

Foundational Principle

Autonomous ecosystems depend on governed compatibility. Operational diversity becomes sustainable only when autonomous systems remain compatible, interoperable, and continuously governable.

Canonical Closing Statement

Autonomous Compatibility establishes the governance conditions required for autonomous systems to operate together across diverse operational environments. By governing compatibility itself, ACPS enables trustworthy interoperability, coordinated autonomous ecosystems, and resilient autonomous operations throughout the complete operational lifecycle.

Module Architecture

→ Operational Interoperability Module (OIM)

→ Protocol Compatibility Module (PCM)

→ Cross-System Trust Module (CSTM)

→ Governance Compatibility Module (GCM)

→ Compatibility Validation Module (CVM)

Related Documents

→ About Autonomous Compatibility Standard (ACPS)

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Canonical Source

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